

Connectivity Community of Practice for Applied Internet Measurement



Advancing School Connectivity Measurement

Measurement Lab (M-Lab) × Giga (UNICEF&ITU)

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Connectivity CoP Research Agenda

[Initial [research agenda](#), as presented in [CoP launch](#), March 2026]

Workstreams:

1. Improving measurement tools
2. Characterization of facility-based connectivity
3. **Internet Quality Barometer for Education (“IQB - Edu”)**
4. New insights from existing measurement data
5. Open topic - BYOI (Bring Your Own Ideas)!

IQB

Internet Quality Barometer (IQB)

Speed tests today

	Download	58.74 Mb/s
	Upload	7.14 Mb/s
	Latency	66 ms
	Retransmission	11.40%



What users want to know

Is my connection good enough for?

Web streaming	
Video streaming	
Video conferencing	
Audio streaming	
Online backup	
Gaming	

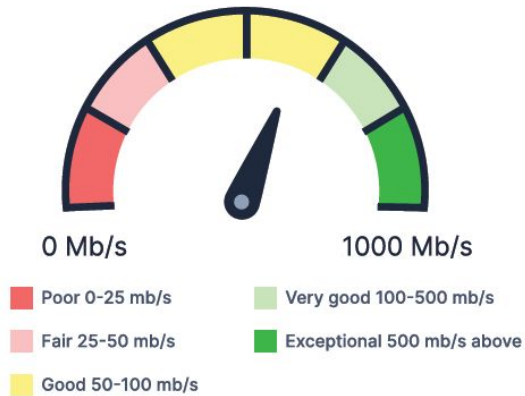
Internet Quality Barometer (IQB)

- What is IQB?
 - A framework to assess Internet Quality from speed test data.
- Why?
 - Translate *technical metrics* (throughput, latency, etc.) to *actionable insights*
- How?
 - Move beyond “speed tests”
 - Calculate a **score** to characterize Internet Quality → from 0 (low quality) to 1 (high quality)

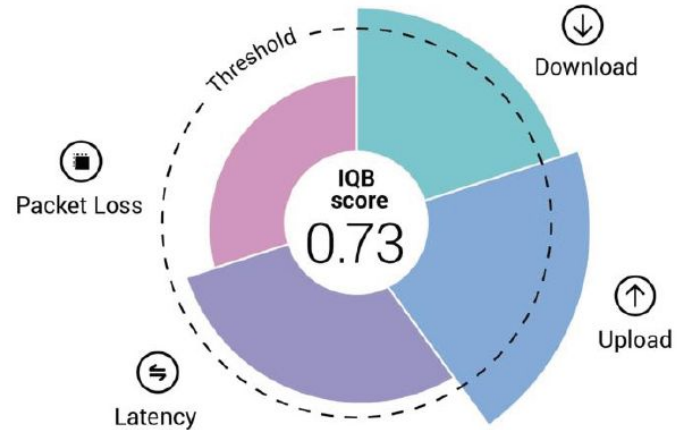
→ Learn more about IQB at <https://www.measurementlab.net/blog/iqb/>

Internet Quality Barometer (IQB)

Speed test



IQB Score



Internet Quality Barometer (IQB)

- From network metrics to use cases
 - Network requirement *thresholds* for minimum & high quality per use case

Use case	Download Throughput		Upload Throughput		Latency		Packet Loss	
	for min quality	for high quality	for min quality	for high quality	for min quality	for high quality	for min quality	for high quality
Web Browsing	10Mb/s	100Mb/s	10Mb/s	Other	100 ms	50ms	1%	0.5%
Video Streaming	25Mb/s	50-100 Mb/s	10Mb/s	10 Mb/s	100 ms	50ms	1%	0.1%
Video Conferencing	10Mb/s	100Mb/s	25Mb/s	100Mb/s	50 ms	20ms	0.5%	0.1%

Use Case Weights

Adjust the importance of each use case (0.0 to 1.0)

Web Browsing



Video Streaming



Audio Streaming



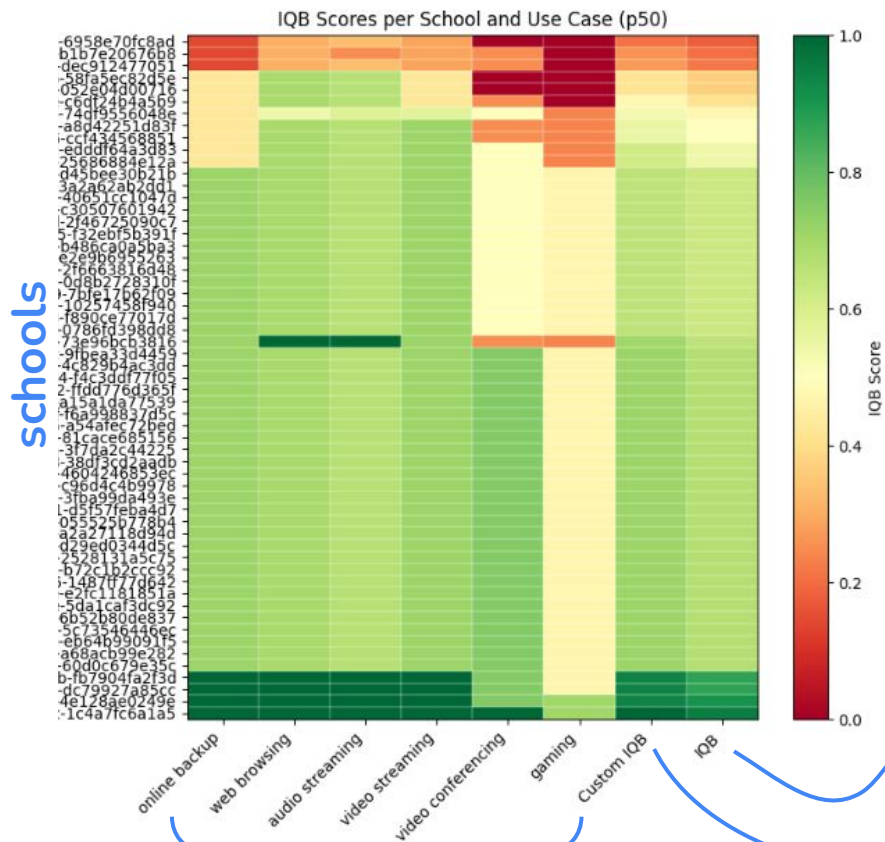
- From use cases to overall user experience
 - Use case weights for different setups (home users, students, etc.)

IQB-Edu

IQB for Education (IQB-Edu)

- **Goal:** design the IQB-Edu framework
 - What use cases matter for pedagogical needs?
→ select use cases & weights
 - What are the network requirements for facilities (vs individual users)
→ update IQB parameters
 - Do IQB scores reflect students experience?
→ extend IQB with other types of measurements
- **Use case:** schools in Moldova
 - *GigaMeter* deployed in schools in Moldova
 - 4 speed test measurements (ndt7) per school per day
 - Data collection & aggregation per school
 - Calculation of IQB scores per school

IQB-Edu for Moldova (v1)



IQB

IQB-Edu

Some key (preliminary) analysis & findings

- Data aggregation (of network metrics per school)
 - **50th** percentile (median) → **average performance**
 - **95th** percentile → best performance / **connection capacity**
- IQB scores
 - 40% (for video conferencing) and 50% (for other use cases) of schools have a perfect IQB score
 - video conferencing is the most demanding use case
- Speed tests (i.e., raw network metrics) vs IQB scores
 - all network metrics affect the IQB scores
 - capacity (95th percentile) is mostly correlated to packet loss
 - average performance (50th percentile) is correlated to all metrics

Call to action

- Code & documentation available on CoP's [GitHub repo](#)
- Check the IQB-Edu [notebook](#) & try it yourself
 - Ideas for more analyses?
 - Test different configurations of IQB-Edu → build your own IQB-Edu!
 - Apply it to other use cases (e.g., data from other schools or facilities)
- Provide recommendations for fine-tuning IQB parameters
 - E.g., play with the data and parameters (prototype), and see what parameters give the correct or wrong insights for you / your use case
 - How to connect parameters to use cases? What policymakers want to consider?
 - E.g., higher percentiles indicate infrastructure capacity (best case performance) vs. lower percentiles relate to population coverage?
- Share list of research questions you consider important to explore
 - and/or visualizations you would like to see

Thank you!

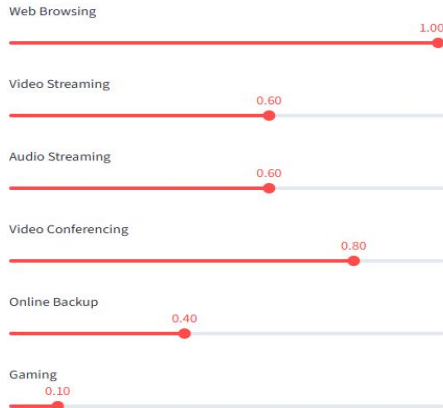
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Audio Streaming	10Mb/s	50Mb/s	10Mb/s	50Mb/s	100 ms	50ms	1%	0.1%
Online backup	10Mb/s	10Mb/s	25Mb/s	200Mb/s	100 ms	100ms	1%	0.1%
Gaming	10Mb/s	100Mb/s	10Mb/s	Other	100 ms	50ms	1%	0.5%

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